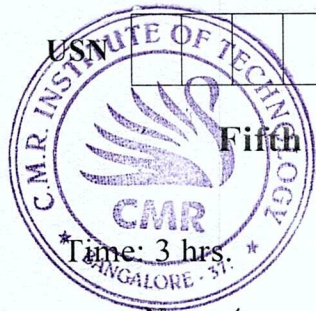




Time: 3 hrs.

CMR

Fifth Semester B.E. Degree Examination, June/July 2025

Electromagnetic Waves

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. A charge of 1C is at (2, 0, 0). What charge must be placed at (-2, 0, 0) which will make y component of total $\vec{E}$ zero at the point (1, 2, 2). (08 Marks)
- b. Explain the types of charge distributions. (06 Marks)
- c. Find the total charge inside a volume having volume charge density as $10z^2 e^{-0.1x} \sin \pi y \text{ C/m}^3$. The volume is defined between $-2 \leq x \leq 2$, $0 \leq y \leq 1$ and $3 \leq z$. (06 Marks)

OR

- 2 a. Explain the properties of electric flux lines. (06 Marks)
- b. Calculate the total charge within the indicated volume $0 \leq \rho \leq 0.1$, $0 \leq \phi \leq \pi$, $2 \leq z \leq 4$ and $\rho_v = \rho^2 z^2 \sin 0.6 \phi$ (08 Marks)
- c. Derive an expression for electric field intensity due to n number of charges. (06 Marks)

Module-2

- 3 a. Find the divergence of $\vec{A}$ at $P(5, \pi/2, 1)$ where $\vec{A} = rz \sin \phi \vec{a}_r + 3rz^2 \cos \phi \vec{a}_\phi$ (08 Marks)
- b. Derive divergence theorem. (06 Marks)
- c. Explain Gauss Law of state one of its applications. (06 Marks)

OR

- 4 a. Given $\vec{A} = 2xy\vec{a}_x + z\vec{a}_y + yz^2\vec{a}_z$, find $\nabla \cdot \vec{A}$ at $P(2, -1, 3)$. (08 Marks)
- b. Determine an expression for volume charge density associated with D

$$D = \frac{4xy}{z} \vec{a}_x + \frac{2x^2}{z} \vec{a}_y - \frac{2x^2y}{z^2} \vec{a}_z$$
 (06 Marks)
- c. State and explain Gauss law in point form. (06 Marks)

Module-3

- 5 a. Solve the Laplace equation for the potential field in the homogeneous region between the two concentric conducting spheres with radii a and b such that $b > a$ if potential $v = 0$ at $r = b$ and $v = v_0$ at $r = a$. Also find the capacitance between two concentric spheres. (10 Marks)
- b. Determine whether the following potential fields satisfy the Laplace's Equation.
 i) $V = x^2 - y^2 + z^2$ ii) $V = r \cos \phi + z$ (10 Marks)

OR

- 6 a. State and prove uniqueness Theorem. (10 Marks)
- b. Given $\vec{H} = 2.5\vec{a}_\theta + 5\vec{a}_\phi$ in spherical coordinate. Find the curl of $\vec{H}$ at $(2, \pi/6, 0)$. (10 Marks)

Module-4

- 7 a. Derive an expression for force on a differential current element. (10 Marks)
- b. Find the magnetization in a magnetic material where :
 i) $\mu = 1.8 \times 10^{-5} \text{ H/m}$
 ii) $\mu_r = 22$ there are 8.3×10^{28} atoms/ m^3 and each atom has a dipole moment of $4.5 \times 10^{-27} \text{ A.m}^2$. (10 Marks)

OR

- 8 a. Obtain magnetic boundary conditions. (10 Marks)
- b. Given a ferrite material which operates in a linear mode with $B = 0.05 \text{ T}$. Calculate the values for magnetic susceptibility, magnetization and magnetic field intensity. Given $\mu_r = 50$. (10 Marks)

Module-5

- 9 a. Wet marshy soil is characterized by $\sigma = 10^{-2} \text{ S/m}$, $\epsilon_r = 15$, $\mu_r = 1$. At frequencies 60 Hz, 100 MHz, 10 GHz indicate whether the soil can be considered as a conductor or dielectric. (08 Marks)
- b. A 160 MHz plane wave penetrates through aluminium of conductivity 10^5 mho/m , $\epsilon_r = 1$, $\mu_r = 1$. Calculate skin depth and also depth at which the wave amplitude decreases to 13.5% of its initial value. (06 Marks)
- c. Derive the expression for modified Amper's circuit law. (06 Marks)

OR

- 10 a. State and prove Poynting Theorem. (10 Marks)
- b. In free space $\vec{E} = 150 \sin(\omega t - \beta z) \vec{a}_x \text{ V/m}$. Calculate the total power passing through a rectangular area of sides 30 mm and 15 mm in $x = 0$ plane. Assume $\frac{E_m}{H_m} = \eta_0$ and $\eta_0 = 120\pi \Omega$. (10 Marks)